

Database Practicum

WEEK 8 ASSIGNMENT



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I. Student Activity

Buatlah tabel untuk sistem perpustakaan yang terdiri dari tiga tabel yaitu books, user, dan flow

dengan kolom sebagai berikut:

1. Books: bookID, bookTitle, authorName, borrowedStatus
2. Flow: flowID, userIDBorrowing, bookIDBorrowed, borrowDate, returnDate
3. User: userID, userName, numberOfBorrowing, numberOfReturning

```
MariaDB [(none)]> use library;
Database changed
MariaDB [library]> CREATE TABLE books (
  ->   bookID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
  ->   bookTitle VARCHAR(100),
  ->   authorName VARCHAR(100),
  ->   borrowedStatus BOOLEAN DEFAULT FALSE
  -> );
Query OK, 0 rows affected (0.031 sec)

MariaDB [library]> CREATE TABLE user (
  ->   userID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
  ->   userName VARCHAR(100),
  ->   numberOfBorrowing INT,
  ->   numberOfReturning INT
  -> );
Query OK, 0 rows affected (0.020 sec)

MariaDB [library]> CREATE TABLE flow (
  ->   flowID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
  ->   userIDBorrowing INT,
  ->   bookIDBorrowed INT,
  ->   borrowDate DATE,
  ->   returnDate DATE,
  ->   FOREIGN KEY (userIDBorrowing) REFERENCES user(userID),
  ->   FOREIGN KEY (bookIDBorrowed) REFERENCES books(bookID)
  -> );
Query OK, 0 rows affected (0.040 sec)
```

The tables

```
MariaDB [library]> INSERT INTO user (userID, userName, numberOfBorrowing, numberOfReturning) VALUES
  -> (1, 'Farrel', 2, 1),
  -> (2, 'Tsaqif', 1, 1),
  -> (3, 'Anindyo', 0, 0);
Query OK, 3 rows affected (0.008 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```

MariaDB [library]> INSERT INTO books (bookID, bookTitle, authorName, borrowedStatus) VALUES
-> (101, 'The Great Gatsby', 'F. Scott Fitzgerald', FALSE),
-> (102, 'To Kill a Mockingbird', 'Harper Lee', TRUE),
-> (103, '1984', 'George Orwell', FALSE),
-> (104, 'Pride and Prejudice', 'Jane Austen', TRUE),
-> (105, 'The Catcher in the Rye', 'J.D. Salinger', FALSE);
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [library]> INSERT INTO flow (flowID, userIDBorrowing, bookIDBorrowed, borrowDate, returnDate) VALUES
-> (1, 1, 102, '2025-10-26', NULL),
-> (2, 2, 104, '2025-10-25', NULL),
-> (3, 1, 101, '2025-10-20', '2025-10-23'),
-> (4, 2, 103, '2025-10-18', '2025-10-21'),
-> (5, 1, 105, '2025-10-10', '2025-10-15');
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

The content of the tables

Dan gunakan subquery untuk

1. Menampilkan semua judul buku yang mempunyai status dipinjam dan tanggal meminjamnya kemarin

```

MariaDB [library]> select bookTitle
-> from books
-> where borrowedStatus = TRUE
-> AND bookID IN (
-> select bookIDBorrowed
-> from flow
-> where borrowDate = '2025-10-26'
-> );
+-----+
| bookTitle                |
+-----+
| To Kill a Mockingbird    |
+-----+
1 row in set (0.001 sec)

```

2. Menampilkan semua judul buku baik buku yang tidak dipinjam dan userID peminjam untuk buku yang dipinjam.

```

MariaDB [library]> SELECT bookTitle,
-> (
-> SELECT userIDBorrowing
-> FROM flow
-> WHERE flow.bookIDBorrowed = books.bookID
-> AND returnDate IS NULL
-> LIMIT 1
-> ) AS userID_borrowing
-> FROM books;

```

bookTitle	userID_borrowing
The Great Gatsby	NULL
To Kill a Mockingbird	1
1984	NULL
Pride and Prejudice	2
The Catcher in the Rye	NULL

5 rows in set (0.002 sec)

3. Menampilkan semua buku yang dipinjam dan semua userID baik dia sebagai peminjaman atau tidak.

```

MariaDB [library]> SELECT u.userID, b.bookTitle
-> FROM user u
-> LEFT JOIN flow f ON u.userID = f.userIDBorrowing
-> LEFT JOIN books b ON f.bookIDBorrowed = b.bookID
-> WHERE b.bookID IN (
-> SELECT bookIDBorrowed
-> FROM flow
-> WHERE returnDate IS NULL) OR b.bookID IS NULL;

```

userID	bookTitle
1	To Kill a Mockingbird
2	Pride and Prejudice
3	NULL

3 rows in set (0.002 sec)

4. Menggunakan satu query, tampilkan daftar semua judul buku dan

userName yang meminjam buku dan user yang meminjam lebih dari 3 buku.

```
MariaDB [library]> SELECT b.bookTitle, u.userName,  
-> CASE  
-> WHEN (  
-> SELECT COUNT(*)  
-> FROM flow f2  
-> WHERE f2.userIDBorrowing = u.userID  
-> ) > 3 THEN 'YA'  
-> ELSE NULL  
-> END AS meminjamLebihDariTiga  
-> FROM books b  
-> JOIN flow f ON b.bookID = f.bookIDBorrowed  
-> JOIN user u ON f.userIDBorrowing = u.userID;
```

bookTitle	userName	meminjamLebihDariTiga
To Kill a Mockingbird	Farrel	NULL
Pride and Prejudice	Tsaqif	NULL
The Great Gatsby	Farrel	NULL
1984	Tsaqif	NULL
The Catcher in the Rye	Farrel	NULL

5 rows in set (0.015 sec)